

Validation Execution Deviation & Exception Governance Module (VEDEGM)

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Governed Space

Validation Execution Deviation & Exception Governance

Canonical Definition

Validation Execution Deviation & Exception Governance Module

(VEDEGM) defines the governance framework for identifying, classifying, assessing, authorizing, containing, correcting, escalating, documenting, and resolving material departures, anomalies, interruptions, substitutions, control losses, unexpected conditions, execution failures, and other exceptions occurring during a Validation Execution Instance.

It establishes the conditions under which differences between the **Authorized Validation Execution Basis** and **Actual Validation Execution** become explicit, traceable, and governable execution events rather than hidden methodological drift.

Operational Role

VEDEGM serves as the fourth internal governance space of the Validation Execution Standard.

The preceding modules establish:

VEBAM — What basis authorizes validation execution?

VEICM — How is that basis instantiated and controlled?

VEOTM — What actually occurred, and can execution be reconstructed?

VEDEGM asks:

Where did actual execution depart from the governed basis, what exceptions occurred, how material were they, and what governance response is required?

The governed progression is:

**Observed Execution → Difference Detection → Deviation Identification
→ Exception Classification → Materiality Assessment → Authorization
Assessment → Impact Assessment → Response & Escalation → Resolution
→ Execution Deviation Record** VEDEGM does not issue the final
Execution Conformity Determination.

It establishes the governed deviation and exception record upon
which that determination depends.

Module Operational Space

VEDEGM governs:

- execution deviations,
- execution exceptions,
- anomaly detection,
- execution divergence,
- execution drift,
- procedural deviation,
- sequence deviation,
- scope deviation,
- object deviation,
- configuration deviation,
- evidence substitution,
- source substitution,
- method-step substitution,
- instrument substitution,
- participant substitution,
- environment deviation,
- access deviation,
- control loss,
- control degradation,
- control bypass,
- interruption,
- execution failure,
- unexpected condition,
- deviation authorization,
- exception authorization,
- deviation materiality,

- exception materiality,
 - deviation impact,
 - exception impact,
 - deviation containment,
 - corrective execution,
 - compensating controls,
 - escalation,
 - resolution,
 - unresolved deviations,
 - cumulative deviation,
 - deviation recurrence,
 - deviation closure,
 - and deviation-and-exception traceability.
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Module Function

VEDEGM applies whenever actual validation execution differs materially or potentially materially from the approved Execution Basis or expected execution conditions.

Its function is to prevent:

- deviations remaining undocumented,
 - execution drift being normalized,
 - unauthorized substitutions becoming invisible,
 - interruptions being treated as irrelevant,
 - failed validation activities being silently omitted,
 - control losses being forgotten after recovery,
 - deviations being accepted without impact assessment,
 - authorized changes being treated as methodologically harmless automatically,
 - repeated minor deviations accumulating without governance,
 - anomalous outputs being ignored because the final result appears favorable,
 - and execution exceptions being converted directly into Validation Object failure or success.
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Deviation and Exception Are Related but Distinct

VALIDOS® distinguishes:

Execution Deviation

from:

Execution Exception

An **Execution Deviation** is a departure from an approved, expected, required, or controlled execution condition.

An **Execution Exception** is a material event, anomaly, failure, interruption, unexpected state, or condition requiring governance attention whether or not it represents a formal departure from the approved method.

For example:

A different instrument than the one approved may be a deviation.

An unexpected system crash during an approved test may be an exception.

An event may be both.

Deviation Detection

Deviation governance begins with the ability to compare:

Authorized Execution Basis

against:

Observed Actual Execution

Relevant signals may come from:

- execution records,
 - control-state logs,
 - operator observations,
 - automated monitoring,
 - object-state changes,
 - configuration snapshots,
 - evidence-use records,
 - source-use records,
 - method-step traceability,
 - or another execution artifact.
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Deviation Identity

A material deviation should be sufficiently identifiable.

A Deviation Record may establish:

- Deviation Identifier,
 - Execution Instance,
 - affected activity,
 - affected criterion,
 - expected condition,
 - actual condition,
 - time,
 - actor or system,
 - affected evidence,
 - affected source,
 - affected method step,
 - affected control,
 - initial classification,
 - and current status.
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Deviation Types

VEDEGM may distinguish different deviation types.

Object Deviation

The Validation Object, version, configuration, or state differs from the authorized basis.

Scope Deviation

Execution moves outside, omits, narrows, or otherwise alters the approved Validation Scope.

Method Deviation

Actual execution departs from the approved Validation Method.

Sequence Deviation

Required validation activities occur in a materially different order.

Configuration Deviation

A system, model, instrument, environment, or validation configuration changes materially.

Evidence Deviation

Different, missing, expired, altered, or unqualified evidence is used.

Source Deviation

A source changes, becomes unavailable, exceeds its approved role, or loses a required reliability condition.

Participant Deviation

A material execution role is performed by a different or unauthorized participant or system.

Control Deviation

A required execution control becomes inactive, degraded, bypassed, or unavailable.

Environment Deviation

Execution occurs under conditions materially different from those required.

These categories may overlap.

Execution Exception Types

Execution exceptions may include:

- system failure,
- instrument failure,
- missing data,
- missing evidence,
- infrastructure outage,
- safety event,
- control failure,
- unexpected system behavior,
- extreme observation,
- contradictory output,
- anomalous measurement,
- operator error,
- AI failure,
- autonomous-system escalation,
- resource failure,
- or another unplanned condition.

An exception may require governance even where no method rule was formally violated.

Deviation Materiality

Not every deviation has equal significance.

VEDEGM may classify deviation materiality as:

Immaterial

Minor

Material

Critical

Execution-Invalidating

Materiality should be assessed relative to:

- Validation Purpose,
- Validation Scope,
- Validation Depth,
- affected Validation Criteria,
- affected method step,
- evidence criticality,
- source criticality,
- consequence of error,

- recoverability,
- and impact on later Validation Determination.

Immaterial Deviation

An Immaterial Deviation has no reasonable material effect on the legitimacy, scope, interpretation, or reconstructability of validation execution.

It may still be recorded where required for traceability.

Minor Deviation

A Minor Deviation represents a limited departure that does not materially undermine the validation but may require documentation, correction, or additional review.

Material Deviation

A Material Deviation can reasonably affect:

- methodological interpretation,
- criterion evaluation,
- evidence use,
- execution completeness,
- execution conformity,
- or permissible Validation Determination.

Material deviations require explicit impact governance.

Critical Deviation

A Critical Deviation materially threatens the legitimacy of one or more critical validation activities.

It may require:

- execution pause,
- re-execution,
- scope restriction,
- escalation,
- method reassessment,
- or Determination Block.

Execution-Invalidating Deviation

An Execution-Invalidating Deviation prevents affected execution activity—or in severe cases the complete execution— from legitimately supporting the intended Validation Determination.

Invalidation should remain bounded to the affected execution scope where possible.

Exception Materiality

Exceptions may also require materiality classification.

An unexpected event can be highly significant even if it does not violate an explicit execution rule.

For example:

A never-before-observed autonomous behavior during a properly executed safety test may be methodologically important despite perfect execution conformity.

VEDEGM therefore governs both rule departure and unexpected operational significance.

Authorized Deviation

A deviation may be authorized before or during execution.

Authorization should identify:

- the proposed deviation,
 - rationale,
 - authority,
 - affected execution activities,
 - affected criteria,
 - expected methodological effect,
 - compensating controls,
 - permitted duration,
 - and applicable conditions.
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Pre-Authorized Deviation

Some Validation Execution Bases may anticipate defined alternative paths.

For example:

If Instrument A becomes unavailable, Instrument B may be used if Calibration Condition X is satisfied.

Where these alternatives are established beforehand, the resulting change may be treated as a governed Pre-Authorized Deviation.

Post-Event Authorization

A deviation discovered after it occurred cannot be made methodologically harmless merely by approving it retrospectively.

Post-event governance may determine whether the deviation is:

- acceptable,
- correctable,
- conditionally usable,
- requiring repetition,
- or invalidating.

VALIDOS® therefore establishes:

Retrospective Authorization ≠ Erasure of Deviation

Unauthorized Deviation

An Unauthorized Deviation occurs where actual execution departs materially from the governed basis without applicable authorization.

This may require:

- immediate correction,
- pause,
- escalation,
- repetition,
- scope restriction,
- or execution invalidation.

Deviation Cause

VEDEGM may identify the cause of a deviation.

Possible causes include:

- human error,
- system error,
- software defect,
- resource limitation,
- environmental change,
- unavailable evidence,
- source failure,
- configuration drift,
- unclear instructions,
- methodological ambiguity,
- external event,
- safety intervention,
- unauthorized action,
- or deliberate modification.

Cause identification can inform corrective and preventive governance.

Deviation Impact Assessment

A Deviation Impact Assessment asks:

- Which execution activity was affected?
 - Which Validation Criterion was affected?
 - Which evidence was affected?
 - Which source conditions changed?
 - Did the deviation affect method validity?
 - Did it affect object state?
 - Did it affect scope?
 - Did it affect sequence?
 - Did it affect traceability?
 - Can affected activity be isolated?
 - Can the effect be corrected?
 - Does re-execution become necessary?
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Localized Impact

Some deviations affect only one narrow execution activity.

Where the affected portion can be reliably isolated, VALIDOS® should avoid invalidating unaffected execution unnecessarily.

This supports proportionate governance.

Systemic Impact

A deviation may have systemic effects where it influences:

- multiple method steps,
- multiple criteria,
- the complete evidence basis,
- the object state,
- the execution environment,
- or the validity of the complete execution sequence.

Systemic deviations may require wider re-execution or reassessment.

Cumulative Deviation

Several small deviations may collectively create a material execution problem.

VALIDOS® therefore establishes:

Minor Deviation + Minor Deviation + Minor Deviation ≠ Necessarily Minor Overall

VEDEGM may assess cumulative effect across an Execution Instance.

Deviation Recurrence

Repeated occurrence of the same deviation may indicate:

- weak control,
- flawed procedure,
- inadequate training,
- unstable environment,
- unsuitable method,
- or systematic execution drift.

Recurrence may increase materiality even when individual instances appear minor.

Execution Drift

Execution Drift is a progressive departure from the approved Execution Basis rather than one discrete event.

Examples include:

- incremental parameter changes,
- repeated informal exceptions,
- expanding access,
- shifting scope,
- changing evidence sources,
- evolving configurations,
- or altered participant roles.

VEDEGM governs accumulated drift as a material execution condition.

Evidence Substitution

Evidence Substitution occurs where approved evidence is replaced by another evidence item or body.

VEDEGM determines whether the substitute is:

- already qualified,
 - equivalent,
 - conditionally acceptable,
 - requiring new Evidence Fitness assessment,
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- or unacceptable.

Substitution must not silently inherit the fitness status of the replaced evidence.

Source Substitution

Where a source changes, the replacement source must not automatically inherit the reliability status of the original source.

A material source substitution may require:

- new identity assessment,
 - provenance assessment,
 - capability assessment,
 - independence assessment,
 - or complete Source Reliability reassessment.
-

Instrument Substitution

An alternative instrument may differ in:

- range,
- calibration,
- precision,
- resolution,
- method,
- software,
- or operating conditions.

Instrument substitution therefore requires equivalence assessment where material.

Participant Substitution

A substitute participant may require assessment of:

- role authority,
- competence,
- independence,
- access,
- and execution permissions.

The presence of another qualified person does not automatically establish role equivalence.

Environment Exception

Unexpected environmental conditions may materially affect validation.

Examples include:

- temperature change,
- network instability,
- load variation,
- system outage,
- unusual user conditions,
- or external interference.

VEDEGM determines whether execution may continue and how affected activity must be treated.

Control Loss

A required control may become:

Degraded

Lost

Bypassed

or:

Unavailable

VEDEGM evaluates:

- when the loss occurred,
 - which activities were affected,
 - whether control recovery occurred,
 - whether affected outputs remain usable,
 - and whether re-execution is necessary.
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Control Bypass

Control Bypass occurs where a required control is intentionally or unintentionally circumvented.

Where the bypass is unauthorized or materially affects execution integrity, it may constitute a critical deviation.

Execution Interruption

An interruption stops or materially disrupts execution.

VEDEGM governs:

- interruption cause,
- duration,
- execution state,
- object-state effect,
- evidence continuity,
- control continuity,
- whether execution can resume,
- and whether previous activity remains valid.

Execution Resume Assessment

Before execution resumes after a material interruption, VEDEGM may require assessment of:

- object continuity,
- configuration continuity,
- environment continuity,
- control restoration,
- evidence continuity,
- source continuity,
- and method-state continuity.

A restart from an altered state may not be methodologically equivalent to uninterrupted execution.

Execution Failure

An Execution Failure occurs where a required validation activity cannot be completed sufficiently.

Examples include:

- test cannot run,
- required evidence unavailable,
- instrument failure,
- model crashes,
- system inaccessible,
- required source unavailable,
- or procedure cannot be completed.

VALIDOS® establishes:

Execution Failure ≠ Validation Object Failure

The failure concerns the validation process.

Corrective Execution

A deviation or exception may permit corrective action.

Corrective execution may include:

- repeating a step,
 - replacing an invalid measurement,
 - restoring a control,
 - correcting configuration,
 - rerunning a test,
 - replacing a participant,
 - restoring evidence continuity,
 - or another governed correction.
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Corrective Action Does Not Erase History

Where a deviation is corrected, the original event remains part of the execution history.

VALIDOS® establishes:

Correction ≠ Historical Deletion

The execution record should preserve:

**Original Condition → Deviation → Corrective Action →
Corrected Condition**

Compensating Control

Where a deviation cannot be removed, a compensating control may reduce its methodological impact.

Examples include:

- independent corroboration,
- additional testing,
- enhanced monitoring,
- reduced Validation Scope,
- additional review,
- duplicated measurement,
- or stricter reliance restrictions.

A compensating control must be proportionate to the deviation.

Deviation Escalation

Material deviations may require escalation where they exceed the authority or capability of the execution actor.

Escalation may involve:

- validation supervisor,
- methodology authority,
- governance authority,
- independent reviewer,
- safety authority,
- or another defined role.

VEDEGM does not define universal organizational hierarchies.

It requires material escalation pathways to remain governed.

Exception Escalation Trigger

Triggers may include:

- critical object change,
 - critical control loss,
 - unknown evidence substitution,
 - source compromise,
 - repeated deviation,
 - safety event,
 - high-consequence anomaly,
 - unauthorized scope expansion,
 - or inability to determine execution impact.
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Deviation Resolution

A deviation may be resolved through:

Accepted as Immaterial

Corrected

Compensated

Re-Executed

Scope Restricted

Conditionally Accepted

Escalated

Execution Portion Invalidated

or:

Unresolved

Resolution must remain traceable to the impact assessment.

Unresolved Deviation

An unresolved material deviation must not disappear simply because execution ends.

It should propagate into:

- Execution Completeness assessment,
 - Execution Conformity assessment,
 - Determination Readiness,
 - or later Validation Determination limitations.
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Deviation Closure

A deviation may be closed when sufficient information exists to establish:

- what occurred,
- why it occurred,
- what it affected,
- what response occurred,
- whether correction or control was sufficient,
- and what residual limitation remains.

Closure does not necessarily mean the deviation became immaterial.

Deviation Status

VEDEGM may establish statuses such as:

Open

Under Assessment

Contained

Corrective Action Required

Corrected

Conditionally Accepted

Escalated

Resolved

Closed with Residual Limitation

or:

Execution-Invalidating

Exception Status

An exception may be:

Observed

Under Investigation

Controlled

Resolved

Escalated

Unresolved

or:

Critical

These states support execution governance without becoming Validation Object states.

Deviation and Traceability

Deviation governance depends upon VEOTM traceability.

A deviation that cannot be sufficiently reconstructed may itself create a Traceability Gap.

Conversely, strong traceability may allow a deviation's effect to be isolated and preserve unaffected validation activity.

Deviation and Execution Conformity

VEDEGM does not itself issue final Execution Conformity.

It provides the structured input needed to determine:

- which deviations occurred,
- their materiality,
- whether they were corrected,
- what remains unresolved,
- and whether execution can still legitimately support later determination.

Deviation Is Not Validation Determination

A material deviation may weaken or invalidate execution.

It does not directly determine whether the Validation Object itself satisfies or fails a Validation Criterion.

VALIDOS® preserves:

Execution Deviation ≠ Object Invalidity

Minimum Implementation Framework

1. Detect Deviation or Exception

Identify actual execution conditions differing from the governed basis or requiring exceptional governance attention.

2. Create Deviation / Exception Record

Record the affected execution activity, expected condition, actual condition, time, actor, object state, and relevant relationships.

3. Classify Type

Determine whether the event concerns:

- object,
- scope,
- method,
- sequence,
- evidence,
- source,
- configuration,
- participant,
- environment,
- control,
- interruption,
- failure,
- anomaly,
- or another exception type.

4. Assess Materiality

Classify the event according to its potential methodological impact.

5. Assess Authorization

Determine whether the deviation was:

- pre-authorized,
- authorized during execution,
- retrospectively assessed,
- or unauthorized.

6. Assess Impact

Determine which criteria, methods, evidence, sources, outputs, and execution activities are affected.

7. Establish Response

Determine whether to:

- continue,
- pause,
- correct,
- compensate,
- repeat,
- restrict,
- escalate,
- terminate,
- or invalidate affected activity.

8. Track Resolution

Maintain the event until sufficient resolution or explicit unresolved status is established.

9. Preserve Residual Limitations

Carry unresolved or residual methodological effects forward.

10. Preserve Lifecycle Traceability

Maintain:

- Deviation or Exception Identifier,
- Execution Instance,
- event type,
- expected condition,
- actual condition,
- affected object state,
- affected scope,
- method step,
- criterion,

- evidence,
- source,
- participant,
- control,
- materiality,
- cause,
- authorization status,
- impact,
- response,
- corrective action,
- compensating controls,
- escalation,
- resolution,
- residual limitation,
- status,
- recurrence,
- cumulative impact,
- version,
- and sufficient lifecycle traceability.

Governance Outputs

VEDEGM may produce:

- Validation Execution Deviation Register,
- Validation Execution Exception Register,
- Deviation Classification Record,
- Exception Classification Record,
- Deviation Materiality Assessment,
- Exception Materiality Assessment,
- Deviation Authorization Record,
- Unauthorized Deviation Record,
- Deviation Impact Assessment,
- Exception Impact Assessment,
- Execution Drift Record,
- Evidence Substitution Record,
- Source Substitution Record,
- Instrument Substitution Record,
- Participant Substitution Record,
- Environment Exception Record,
- Control Loss Record,
- Control Bypass Record,
- Execution Interruption Record,
- Execution Failure Record,
- Corrective Execution Record,
- Compensating Control Record,
- Deviation Escalation Record,
- Exception Escalation Record,
- Cumulative Deviation Assessment,
- Deviation Recurrence Record,
- Deviation Resolution Record,
- Residual Limitation Record,
- Deviation Closure Record,
- and Deviation & Exception Traceability Record.

Use Case 1 — AI Model Validation

Scenario

During validation of an AI model, one approved benchmark becomes temporarily unavailable and the execution team replaces it with a similar benchmark.

Application

VEDEGM records the substitution as an Evidence and Method-related Deviation.

The replacement benchmark does not automatically inherit the qualification status of the approved benchmark.

Its effect on:

- criteria coverage,
- Evidence Fitness,
- method comparability,
- and execution validity

is assessed.

Result

Execution may continue only if the substitution is sufficiently governed, or the affected validation activity may require later re-execution.

Use Case 2 — Industrial Validation

Scenario

A required sensor temporarily loses calibration during a critical measurement sequence.

Application

VEDEGM identifies:

- Control Loss,
- affected measurement period,
- affected Validation Criteria,
- affected outputs,
- and time of calibration recovery.

Result

Measurements generated during the affected period remain distinguishable from valid measurements and may require repetition rather than contaminating the complete execution record.

Use Case 3 — Autonomous Validation Agent

Scenario

An autonomous validation agent encounters an unexpected system condition and uses a tool outside the execution path anticipated by the Validation Method.

Application

VEDEGM determines whether the action was within delegated authority, whether it changed the methodological basis, what outputs were affected, and whether human escalation was required.

Result

The event becomes a governed execution exception rather than an invisible autonomous adaptation.

Architectural Position

Validation Execution Deviation & Exception Governance is the fourth internal governance space of the **Validation Execution Standard (VEXS)**.

The progression now becomes:

**Execution Basis & Authorization → Execution Instantiation & Control
→ Execution Observation & Traceability → Execution Deviation & Exception Governance**

VEBAM establishes:

What authorizes execution?

VEICM establishes:

How is execution instantiated and controlled?

VEOTM establishes:

What actually happened?

VEDEGM establishes:

Where did execution differ, what exceptional conditions occurred, and how were those differences governed?

The fifth and final VEXS module can then determine:

Was the overall execution sufficiently complete and conformant to support Validation Determination?

Foundational Principle

Execution deviations do not destroy validation merely by occurring; they destroy validation when their existence, materiality, impact, and resolution are not governable.

Canonical Closing Statement

Validation Execution Deviation & Exception Governance Module establishes the controlled exception layer of VALIDOS® validation execution by governing departures, anomalies, substitutions, interruptions, failures, control losses, unexpected conditions, authorization, materiality, impact, corrective action, compensation, escalation, recurrence, cumulative effects, resolution, residual limitations, and traceability. VEDEGM ensures that operational variation does not disappear into execution history and that neither deviation nor exception is allowed to influence Validation Determination without first becoming an explicit and governable methodological condition.

Parent Resources

→ [VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

→ [VALIDOS® Architecture Map](#)

→ [VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

→ [Previous module: Validation Execution Observation & Traceability Module \(VEOTM\)](#)

→ [Next module: Validation Execution Completeness & Conformity Module \(VECCM\)](#)

Related Documents

→ Validation Execution Standard

→ About Validation Execution Standard

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